MATHS I · WEEK 1

8 set formulas that keep coming back

Sets and Venn diagrams show up in almost every Qualifier paper.

SETS

Counting subsets (|A| = n)

Power set

$$|P(A)| = 2^n$$

Proper subsets

$$2^n - 1$$

Subsets of size k

$$C(n, k) = n! / (k! (n - k)!)$$

Cartesian product

$$|A \times B| = |A| \cdot |B|$$

VENN DIAGRAMS

Counting people in groups

Two sets

$$|A \cup B| = |A| + |B| - |A \cap B|$$

Three sets

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |A \cap C| + |A \cap B \cap C|$$

Neither / none

$$\text{Total} - |A \cup B \cup C|$$

Smallest overlap

$$\max(0, |A| + |B| - \text{Total})$$

METHOD

Fill a Venn diagram from the centre out

1

All three

Put the triple overlap in the middle first.

2

Exactly two

Take the middle away from each pair.

3

Only one

Then fill each 'only' region.

4

None

Total minus the union goes outside.

TRAPS

Where marks are lost

⚠ {∅} is not empty. It has one element.

⚠ {1, 2, 2} = {2, 1}. Repeats and order do not matter.

⚠ 'Both A and B' includes people also in C. 'Only A and B' does not.

⚠ If the total is not given, never assume 'none' is 0.

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Timed mocks on the real exam screen, graded with the official answer keys.

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